DEEPAK KUMAR

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SUMMARY

Results-driven Data Scientist with 5+ years of experience designing scalable machine learning models, optimizing network performance, and analyzing large-scale telemetry data. Proven expertise in time-series forecasting, predictive modeling, and big data processing using Python and Azure Databricks. Adept at translating complex network and IoT telemetry into actionable insights to drive coverage forecasting and infrastructure optimization.

EXPERIENCE

Microsoft

AI/ML Applied Scientist *Dec 2022 – Present*

- Built an enterprise Copilot evaluation framework on Azure ML that automated quality benchmarking across 12 product teams, cutting manual review cycles by 70% and improving response relevance scores by 18% through reinforcement learning from human feedback (RLHF).
- Led end-to-end development of a multi-agent orchestration system using AutoGen and Azure AI Foundry, enabling autonomous task decomposition across 5 enterprise workflows and reducing analyst toil by 40% — deployed to 3,000+ internal users within 6 months.
- Designed a responsible-AI content moderation pipeline integrating Azure Content Safety with a custom fine-tuned Mistral-7B classifier, achieving 97.3% precision on harmful content detection and reducing false-positive escalations by 55% across global markets.
- Optimized distributed model training on Azure Databricks with DeepSpeed ZeRO-3, cutting GPU-hour costs by 35% on 70B-parameter workloads and reducing time-to-deploy for production LLM checkpoints from 5 days to under 18 hours.

Tiger Analytics

Data Scientist *Jun 2022 – Dec 2022*

- Built a Bayesian Market-Mix Model using Python, successfully reallocating a 5M EUR budget and increasing marketing ROI by 10% for MARS Petcare Germany.
- Delivered executive dashboards to guide FY23 investment decisions, partnering with global cross-functional stakeholders to translate complex data insights into actionable strategies.

Verizon

Data Scientist *Aug 2020 – Jun 2022*

- Forecasted power usage and telemetry metrics across 100K cell towers using a Prophet and ARIMA ensemble, informing a USD 800M OPEX budget and improving coverage forecasting accuracy by 18% on hybrid cloud infrastructure.
- Optimized fiber-rollout milestones and network performance by deploying regression models, improving on-time delivery by 25% and reducing manual scheduling effort by 4× across regional engineering teams.

SKILLS

Data & AI Engineering: Python, SQL, MySQL, PostgreSQL, Azure Databricks, PySpark, REST APIs, Data Pipelines, Hybrid Cloud, IoT Telemetry

Machine Learning & MLOps: MLflow, Docker, CI/CD (GitHub Actions), MLOps Pipeline Automation, Production ML Deployment, XGBoost, Random Forest, Network Performance Optimization

GenAI & NLP: Agentic AI, Workflow Automation, Llama-3, Phi-3, Mistral, GPT-4, RAG, LangChain, Vector DB (FAISS/Pinecone), Hugging Face

Time-Series & Analytics: Prophet, ARIMA, Fraud Prevention, Root Cause Analysis, Agile, Mesh Networks, BLE, LoRa

PROJECTS

Local LLM Chat Application — Designed and deployed a privacy-first AI assistant running entirely on-device using quantized Llama-2-7B and Mistral-7B via llama.cpp, achieving sub-2s response latency on CPU-only hardware with zero cloud dependency. Built a multi-session Streamlit UI with conversation history, model switching, and token-budget controls; reduced inference memory footprint by 60% through 4-bit GGUF quantization.

Real-Time Ad-Detection CV Pipeline (US Patent 12126855B2) — Architected and patented a production-grade streaming-commerce engine that identifies branded products in live video frames at 30 FPS using YOLOv5 for object detection and CRAFT for text localization. Pipeline achieved 91% mAP on a proprietary 50K-frame benchmark, reduced manual product-tagging effort by 80%, and enabled real-time shoppable overlays for 3 major OTT platforms — generating an estimated USD 4M in incremental affiliate revenue.

AutoML Image Classifier — Built a no-code AutoML platform with a Streamlit GUI that lets domain experts label images, trigger fine-tuning of a ResNet-50 backbone via PyTorch Lightning, and deploy the model as a REST endpoint — all within a single browser session. Achieved 90%+ top-1 accuracy across 5 production use-cases; cut model iteration cycles from 2 weeks to under 4 hours and eliminated the need for dedicated ML engineers on 3 client teams.

Time-Series Forecasting Application — Developed an ensemble forecasting platform in Python and Streamlit that blends Prophet, ARIMA, and XGBoost predictions using a stacking meta-learner, reducing MAPE by 22% versus any single model. Deployed for network-capacity planning across 50K+ cell-tower sites; interactive dashboards let planners drill into coverage gaps, run what-if scenarios, and export board-ready business plans — directly informing a USD 800M OPEX allocation.

EDUCATION

IIT Hyderabad
MTech in Data Science *Aug 2025 – May 2027*

Chandigarh Engineering College
B.Tech, Computer Science *Aug 2016 – May 2020*

ACHIEVEMENTS

- US Patent 12126855B2: Streaming Commerce real-time ad-detection engine – USPTO (2024)
- Global Make It Happen 2024 Award, Top 200 of 400K+ employees (2024)
- Winner, Provider Ops Data-Science Hackathon (2024)
- 3× Spot Awards for data-science impact – Verizon (2022)